

Key tools that can be used for:

Identity management

- uPort, <https://www.uport.me/>
- Hyperledger Indy, <https://www.hyperledger.org/use/hyperledger-indy>



Figure 3: IPFS platform for decentralised storage

Data integrity and protection

- BigchainDB, <https://www.bigchaindb.com/>
- Ethereum + IPFS (InterPlanetary File System), <https://ipfs.tech/>



Figure 4: BigChainDB decentralised database

IoT security

- Nodle, <https://www.nodle.com/>
- IOTA, <https://www.iota.org/>

Threat intelligence sharing

- IBM Collaborative Threat Intelligence Platform (Open-Source Components), <https://www.ibm.com/security>
- MITRE ATT&CK + Blockchain Frameworks, <https://attack.mitre.org/>

Digital evidence management

- Guardtime, <https://guardtime.com/>
- Hyperledger Fabric, <https://www.hyperledger.org/use/fabric>

and the risk of human mistakes.

The blockchain-based system makes it possible to securely transfer evidence between jurisdictions. In more complicated cross-border investigation situations, it was frequently difficult for law enforcement agencies in different jurisdictions to securely exchange digital evidence. By putting in place a permissioned blockchain network, only authorised agencies can access records of verified evidence.

Blockchain technology has shown its potential to change the paradigm of digital evidence by ensuring security, transparency and reliability. With institutions in the legal and law enforcement fields implementing blockchain based solutions, digital evidence will have greater integrity, which will contribute to fair legal processes.

Cybersecurity is a constantly evolving discipline that is crucial for securing electronic assets against a wide range of cyber-attacks. With technology advancing, the cybercriminals have advanced as well and that is why security measures are also evolving. While modern technologies do improve protection, a multi-layered security approach with traditional security mechanisms should be integrated for enhanced protection.

Training and awareness campaigns are critical to security solutions, particularly as human error continues to be a leading cause of cyber events. More intelligent security frameworks need to emerge — mandated cyber hygiene policies, championed by government and commercial entities, as well as funded research pursuits on academic campuses. A proactive, research-driven approach will remain critical for building a more secure and resilient digital future. **END** 🐧

intermediaries. The decentralised nature of the blockchain means that there is no single point of failure, making it less likely for evidence to be lost or corrupted due to system failures or cyber-attacks.

Another key advantage is the automated legal compliance provided by smart contracts. If a piece of evidence needs to be transferred between departments or presented in

court, the smart contract automatically executes the authorisation process required, ensuring adherence to legal protocols. This improves security as well as decreases administrative loads

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